

● HARD

● INFORMATION AND IDEAS

Inferences

10 questions · ~15 min

02

Q1

INFORMATION AND IDEAS

In 2022, Crystal Reeck and colleagues studied whether the decision-making modes that guide consumers influence their choice between nonenvironmentally friendly standard electricity plans and environmentally friendly green plans that cap electricity usage. Study participants who self-reported using either an Affect Mode or Role Mode—which prioritize choices that have a stronger positive emotional or social impact, respectively—were more likely to select a green plan. Conversely, participants using a Calculation Mode—which aims to minimize both financial cost and personal inconvenience—were more likely to select a standard plan, even when the green option was cheaper. This finding suggests that participants using a Calculation Mode _____blank

Which choice most logically completes the text?

- A. were equally unlikely to factor the financial savings of the green plan into their decision-making as were participants using either the Affect or Role Modes.
- B. may have been less strongly motivated to appear socially responsible with their choice of plan than they realized.
- C. may have determined that the green plan imposed additional burdens on them that were not sufficiently offset by the potential financial savings.
- D. were less likely to believe that the green plan was truly cost-effective than were participants using either the Affect or Role Modes.

Indigenous cultures possess unique knowledge of the medicinal uses of plants. According to a 2021 study, 73 percent of the medicinal uses of plants native to North America are reflected in the vocabulary of a single Indigenous language. However, as more and more Indigenous people exclusively speak a globally dominant language, such as English, their ancestral languages fade from daily use. These facts lend added importance to tribal nations' efforts to preserve their languages. By ensuring the continued use of Cherokee, Ojibwe, and the hundreds of other Indigenous languages in what is now the United States, tribal nations are also _____ blank

Which choice most logically completes the text?

- A. increasing the number of medicinal plants represented in the vocabularies of Indigenous languages.
- B. transmitting terms for medicinal plants from Indigenous languages to globally dominant languages.
- C. preserving knowledge about the medicinal value of plants native to the tribal nations' lands.
- D. ensuring that citizens of tribal nations have physical access to medicinal plants.

Studies conducted in individual Western societies have demonstrated associations between personality traits and five musical factors (mellowness, unpretentiousness, sophistication, intensity, and contemporariness) underlying individual preferences for styles of Western music. To investigate these associations across cultures, David M. Greenberg et al. collected music-preference assessments for Western genres and self-reported personality traits from participants in fifty-three countries across six continents. The study confirmed that the five-factor framework accurately captured participants' tastes in Western music, and, moreover, the study found similar correlations between patterns of these factors and of personality traits, suggesting that _____ blank

Which choice most logically completes the text?

- A. across cultural contexts, people who share similar profiles of personality traits tend to prefer listening to similar types of Western music.
- B. the five-factor framework can likely be used to predict preferences for non-Western music styles based on personality traits even if the characteristics of those styles substantially differ from characteristics of Western music styles.
- C. the strength of the relationship between personality traits and musical preferences varies less across cultures than researchers had previously assumed.
- D. people with a relatively high degree of familiarity with Western music styles are likely to express stronger preferences for those styles than people with a relatively low degree of familiarity with those styles are.

Ratified by more than 90 countries, the Nagoya Protocol is an international agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge of wild plants and animals are utilized by agricultural corporations. However, the protocol has shortcomings. For example, it allows corporations to insist that their agreements with communities to conduct research on the commercial uses of the communities' resources and knowledge remain confidential. Therefore, some Indigenous advocates express concern that the protocol may have the unintended effect of _____ blank

Which choice most logically completes the text?

- A. diminishing the monetary reward that corporations might derive from their agreements with Indigenous communities.
- B. limiting the research that corporations conduct on the resources of the Indigenous communities with which they have signed agreements.
- C. preventing independent observers from determining whether the agreements guarantee equitable compensation for Indigenous communities.
- D. discouraging Indigenous communities from learning new methods for harvesting plants and animals from their corporate partners.

Outi Tervo and team studied the effect of human-caused noise on narwhals (*Monodon monoceros*), arctic marine mammals that are sensitive to acoustic changes in their environment. Hypothesizing that elevated sound levels affect foraging among narwhals, Tervo's team compared narwhal diving behaviors in natural sound conditions with those behaviors in two human-caused sound exposure conditions—ship sounds and ship sounds coupled with sonic pulses. Both exposure conditions resulted in significant decreases in the number and target depth of deep dives (associated with foraging) relative to natural conditions. However, differences between diving behaviors in the two exposure types were negligible, a finding that could be attributed to the fact that _____ blank

Which choice most logically completes the text?

- A. sonic pulses can be heard at significantly greater ocean depths than ship sounds can.
- B. ship sounds contribute so much to the overall sound level that the addition of sonic pulses has little effect on the narwhals' auditory environment.
- C. narwhals forage at shallower depths in the presence of ship sounds alone than in the presence of ship sounds coupled with sonic pulses.
- D. the narwhals weren't as sensitive to human-caused sounds as the researchers had predicted.

Aptamers—synthetic DNA or RNA molecules that bind to target molecules—can be used to test for foodborne bacterial pathogens, though their specificity (the probability of returning a negative result in the absence of the focal pathogen) in real-world foods has been unclear. Sandeep Somvanshi et al. fabricated test paper incorporating aptamers targeting strain O157:H7 of the bacteria *Escherichia coli*; the paper shifts from pink to purple as the aptamers bind to target molecules. Somvanshi et al. tested the paper in store-bought pear juice they treated with *E. coli* O157:H7, other strains of *E. coli*, or other bacteria species. Following exposure, the paper from the O157:H7 test was purple while papers from the other tests were pink, suggesting that _____ blank

Which choice most logically completes the text?

- A. aptamer-based tests in real-world foods are more likely to show a high degree of specificity if the focal pathogen is *E. coli* O157:H7 than if the focal pathogen is another strain of *E. coli* or another species.
- B. uncertainty about the specificity of aptamer-based tests for pathogens in real-world foods may be due to the similarity between *E. coli* O157:H7 and other *E. coli* strains.
- C. the specificity of the tests in a real-world food was unaffected by the aptamers' tendency to bind to different strains of *E. coli*.
- D. the aptamers successfully bound to *E. coli* O157:H7 and the tests displayed a high degree of specificity in a real-world food.

Archaeologists and historians used to believe that the Maya civilization during its Classic period (roughly 250–900) lacked agricultural marketplaces. One reason for this belief was that these scholars misunderstood the ecology of the regions the Maya inhabited. Marketplaces typically emerge because different individuals or groups want to trade resources they control for resources they don't control. Scholars seriously underestimated the ecological diversity of the Maya landscape and thus assumed that _____ blank

Which choice most logically completes the text?

- A. marketplaces likely would not have attracted many traders from outside the regions controlled by the Maya.
- B. farming practices would have been largely the same throughout Maya lands even if the crops people produced varied significantly.
- C. marketplaces would not have enabled Maya people to acquire many products different from those they already produced.
- D. farmers would trade agricultural products only if they had already produced enough to meet their own needs.

The *Mammillaria* cactus *M. boolii* occurs naturally only in the state of Sonora in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like *M. boolii*, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. If this view is correct, the single-species approach to the conservation of *M. boolii* could thus _____ blank

Which choice most logically completes the text?

- A. lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.
- B. allow conservationists to better consider how climatic changes affecting Sonora may reduce the number of species competing with *M. boolii*.
- C. erroneously shift the focus of conservation efforts away from *M. boolii* itself.
- D. fail to consider the ways in which the survival of *M. boolii* may be influenced by changes in the populations of other species that inhabit Sonora.

For its 1974 work *Instant Mural*, the Chicano art collective Asco taped members Patssi Valdez and Humberto Sandoval to an outdoor wall in East Los Angeles. The work is manifestly a commentary on constraint, but many critics focus on Valdez and the social constraints women faced at the time, which is understandable but leaves the presence of Valdez's male collaborator Sandoval unexplained. We should instead consider that in 1974, the art establishment's recognition of Chicano artists was (and had long been) restricted to sociohistorical muralists, leaving nonmuralist Chicano artists—like Asco's members—struggling to even exhibit their work; attending to this context opens an interpretation that accounts for all the evidence, allowing us to conclude that _____ blank

Which choice most logically completes the text?

- A. while Valdez's presence in *Instant Mural* represents the social constraints placed on women at the time, Sandoval's presence represents Chicano muralists' frustration at their lack of recognition by the art establishment.
- B. the main subject of *Instant Mural* is female Chicano artists' experience of being doubly constrained by gender-role expectations and the marginalization of certain types of art.
- C. *Instant Mural* is a reflection on the constraining aesthetic expectations placed on Chicano artists in general rather than on the social constraints placed on women specifically.
- D. *Instant Mural* is best understood not as a critique of the social constraints placed on women but rather as a critique of sociohistorical muralists' depictions of Chicano culture.

The alpaca was domesticated by Indigenous peoples in the Andes about 7,000 years ago. But which wild species did it descend from, the vicuña or guanaco? A research team led by Ruiwen Fan may have solved the mystery, concluding that the alpaca is the domesticated form of the vicuña but that the modern alpaca gets only 64 percent of its genetic material from its wild ancestor. The rest comes from the domesticated llama. The llama, meanwhile, gets 95.5 percent of its genetic material from its own wild ancestor, the guanaco, and the rest from the alpaca. The llama and alpaca apparently interbred widely for only a handful of generations between 400 and 600 years ago. Assuming that the findings of Fan's team are valid, it can be inferred that _____ blank

Which choice most logically completes the text?

- A. modern llama populations have a greater degree of genetic diversity, on average, than modern alpaca populations do.
- B. the domestication process of the alpaca may have involved some introduction of genetic material from the llama.
- C. the period of interbreeding resulted in a greater genetic difference between alpacas and their wild ancestors than between llamas and their wild ancestors.
- D. if they were subjected to genetic testing, modern populations of guanacos and vicuñas would likely show traces of ancient interbreeding as well.